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Structural cell for facilitating tree root growth

Abstract

A structural cell system for supporting hardscape, allowing tree root growth, and managing stormwater underneath the hardscape. The system may include: a base having a plurality of receptacles and a plurality of support members interconnecting the receptacles; a plurality of legs each sized and shaped to be attachable to the base within one of the receptacles so as to extend from the base, and to be attachable to another of the legs so that pairs of legs attached to each other collectively extend from the base; and a top attachable to the legs. Outer edges of the base, the top, and the legs attached thereto define a volume, and are configured to support at least that portion of the hardscape overlying the top as well as a commercial vehicle traffic load thereon, while maintaining soil in a substantially uncompacted state throughout at least approximately ninety percent of the volume.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/410,916 filed on May 13, 2019, which is a continuation of U.S. patent application Ser. No. 15/686,573 filed on Aug. 25, 2017, which is a divisional of U.S. patent application Ser. No. 14/684,214 filed Apr. 10, 2015 and titled "More Efficient Structural Cell For Facilitating Tree Root Growth", which is incorporated herein by reference in its entirety.

BRIEF DESCRIPTION

(1) This invention relates generally to tree growth technology. More specifically, this invention relates to structural cells for more efficiently facilitating tree root growth and water retention.

BACKGROUND

(2) The design of many modern dense urban landscapes often calls for the placement of trees within paved-over areas or areas covered by other hardscapes. In particular, such designs often call for trees to be placed in close proximity to roads, sidewalks, and other load bearing pathways. However, the design and construction methods of these pathways and the loads they carry often compact the soil underneath to such an extent that it is often difficult for tree roots to sufficiently penetrate the soil. As a result, trees planted in close proximity to these hardscapes may not survive or grow to the full extent envisioned.

(3) Various solutions to this problem have been proposed. For example, structural cell systems such as those disclosed in U.S. Pat. Nos. 7,080,480 and 8,065,831, which are both hereby incorporated by reference in their entireties and for all purposes, have been designed to facilitate the growth of trees near hardscapes, while allowing for soil aeration, water drainage, and the like. It is, however, desirable to improve various aspects of such cells. Accordingly, continuing efforts exist to make such structural cells support hardscapes better, while improving the manufacturability and other characteristics of such cells.

SUMMARY OF THE INVENTION

(4) The invention can be implemented in numerous ways. Accordingly, various embodiments of the invention are discussed below.

- (5) In one embodiment, a structural cell system for supporting hardscape and allowing tree root growth and managing stormwater underneath the hardscape comprises: a base having a plurality of receptacles and a plurality of support members interconnecting the receptacles; a plurality of legs each sized and shaped to be attachable to the base within one of the receptacles so as to extend from the base, and to be attachable to another of the legs so that pairs of legs attached to each other collectively extend from the base; and a top attachable to the legs. Outer edges of the base, the top, and the legs attached thereto define a volume, and are configured to support at least that portion of the hardscape overlying the top as well as a commercial vehicle traffic load thereon, while maintaining soil in a substantially uncompacted state throughout at least approximately ninety percent of the volume.
- (6) Outer edges of the receptacles may extend beyond outer edges of the support members.
- (7) The base and the legs may each comprise an unreinforced plastic, such as high density polyethylene (HDPE).
- (8) The base, the top, and the legs attached thereto may be sized and shaped to support a load of at least approximately 15 psi across substantially the entire top.
- (9) At least two of the receptacles may lie along a first direction, and at least two of the receptacles may lie along a second direction different from the first direction. Also, each of the legs may have a cross-sectional shape having protrusions, the protrusions arranged so that, when each of the legs is attached to a receptacle, its protrusions extend in the first and second directions.
- (10) The plurality of legs may comprise legs having a first length and legs having a second length shorter than the first length. Each pair of legs may comprise a leg having the first length and a leg having the second length.
- (11) The base may be configured so that plural ones of the bases are stackable in a vertical direction so as to be substantially coupled in a lateral direction perpendicular to the vertical direction, and so as to be substantially uncoupled in the vertical direction; and the top may be configured so that plural ones of the tops are stackable in the vertical direction so as to be substantially coupled in the lateral direction and substantially uncoupled in the vertical direction. The system may further comprise a bioretention system positioned within the structural cell system; and tree roots extending into the bioretention system, the tree roots being from a tree positioned proximate to the structural cell system.
- (12) The base, the associated legs, and the top may collectively comprise a single structural cell, and the structural cell may have no mechanism for coupling to another one of the structural cells.
- (13) In another embodiment, a method of facilitating the growth of a tree comprises: placing a plurality of structural cells side by side and adjacent to each other, so that the structural cells are substantially uncoupled to each other and so that the structural cells substantially surround the rootball of a tree. Each structural cell may comprise: a base having a plurality of receptacles and a plurality of support members interconnecting the receptacles; a top; and a plurality of legs affixing the base to the top. Outer edges of the base, the top, and the legs affixed thereto define a volume, and are configured to support at least that portion of a hardscape overlying the top as well as a commercial vehicle traffic load thereon, while maintaining soil in a substantially uncompacted state throughout at least approximately ninety percent of the volume.
- (14) The method may further comprise filling the cells with a substantially loosely compacted soil, so as to allow roots of the tree to grow into the soil within the structural cells while simultaneously providing storage and pollutant removal of stormwater.
- (15) Each receptacle may be connected to the top by a pair of interconnected ones of the legs.
- (16) The filling the cells may further comprise filling the cells with material layers so as to form a bioretention system within the cells, each of the material layers being substantially uncompacted so as to allow roots of the tree to grow into the bioretention system within the cells. Other aspects and advantages of the invention will become apparent from the following detailed description taken in

conjunction with the accompanying drawings which illustrate, by way of example, the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a better understanding of the invention, reference should be made to the following detailed description taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 illustrates operation of an exemplary structural cell system of the invention;
- (3) FIG. 2A illustrates an isometric detail view of an assembled structural cell of the invention;
- (4) FIG. 2B is a side view of the assembled structural cell of FIG. 2A;
- (5) FIG. 2C is a cross-section view taken along line I-I' of FIG. 2B;
- (6) FIG. 3A is a plan view of the base of the structural cell of FIG. 2A;
- (7) FIGS. 3B and 3C are side views of the base of FIG. 3A;
- (8) FIG. 4A is an isometric view of one embodiment of a leg of the structural cell of FIG. 2A, where in this embodiment the leg is a longer leg;
- (9) FIG. 4B is a side view of the leg of FIG. 4A;
- (10) FIG. 4C is a cross-section view of a center portion of the leg of FIG. 4A;
- (11) FIG. 5A is an isometric view of another embodiment of the leg of the structural cell of FIG. 2A, where in this embodiment the leg is a shorter leg;
- (12) FIG. 5B is a side view of the leg of FIG. 5A;
- (13) FIG. 5C is a cross-section view of a center portion of the leg of FIG. 5A;
- (14) FIG. 6A is a plan view of the deck of the structural cell of FIG. 2A;
- (15) FIGS. 6B and 6C are side views of the deck of FIG. 6A;
- (16) FIG. 7A is an isometric detail view of an assembled structural cell, using both longer legs and shorter legs;
- (17) FIG. 7B is a side view of the structural cell of FIG. 7A;
- (18) FIG. 7C is a cross-section view taken along line II-II' of FIG. 7B;
- (19) FIG. 8A is an isometric detail view of an assembled structural cell, using only shorter legs;
- (20) FIG. 8B is a side view of the structural cell of FIG. 8A;
- (21) FIG. 8C is a cross-section view taken along line III-III' of FIG. 8B;
- (22) FIG. 9A is an isometric detail view of stacked bases constructed in accordance with embodiments of the present invention;
- (23) FIG. 9B is a side view of the stacked bases of FIG. 9A;
- (24) FIG. 9C is a cross-section view taken along line IV-IV' of FIG. 9B;
- (25) FIG. 10A is an isometric detail view of stacked decks constructed in accordance with embodiments of the present invention;
- (26) FIG. 10B is a side view of the stacked decks of FIG. 10A; and
- (27) FIG. 10C is a cross-section view taken along line V-V' of FIG. 10B.
- (28) Like reference numerals refer to corresponding parts throughout the drawings. The various Figures are not necessarily to scale.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

(29) In one aspect, the invention relates to structural cell systems that are placed beneath hardscape. The cells are strong enough to structurally support the hardscape, effectively bearing its weight along with the weight of any load it carries. Furthermore, even though the cells are strong enough to offer structural support of a hardscape, the cells are also designed to be relatively lightweight and open, allowing approximately 90% of their volume, or more, to be free volume that can contain uncompacted soil, tree roots, utilities, and the like. Some of the components of the cell systems are stackable for better transportability. The cells achieve these attributes through a design that includes

a bottom frame, attachable support members, and an attachable deck or top. The various components of the cells are designed to utilize less material relative to the cells of U.S. Pat. Nos. 7,080,480 and 8,065,831, while also being able to support the load of overlying hardscape and commercial vehicles yet being made of a more inexpensive material. Furthermore, the legs are attachable to each other so that individual cells can be made of multiple sizes, some of which are taller than the individual cells of U.S. Pat. Nos. 7,080,480 and 8,065,831. This allows a single layer of cells to take the place of multiple layers of the previous cells, thus further reducing the amount of material used.

(30) FIG. 1 illustrates an exemplary application of the structural cells of the invention. Here, a tree **10** grows its roots in the soil **50** underneath a hardscape **20** and layer of aggregate **30**. Structural cells **40** are stacked between the hardscape **20** and aggregate **30** above, and foundation **60** below. The cells **40** are sufficiently rigid that they structurally support the weight of the hardscape **20**, aggregate **30**, and any loads above (e.g., cars, pedestrians, etc.), transferring it to the foundation **60** rather than the soil **50**. This maintains the soil **50** within the structural cells **40** in a relatively uncompacted state, allowing roots from the tree **10** to grow therethrough as shown. In addition, the rigidity of the cells **40** allows a relatively small number of support members to bear structural loads. In this manner, the cells **40** maintain a large amount of continuous open volume within, free of excessive numbers of support members that take up space and prevent large tree roots from growing therethrough. Space is left between adjacent structural cells **40** for ease in removing/repairing individual cells **40** without disturbing the rest.

(31) In some embodiments, the structural cells **40** are configured to satisfy a number of constraints. For example, the cells **40** should be composed of a material capable of withstanding an underground environment that can contain water. This material should also be of sufficient strength to support a hardscape **20**, aggregate **30**, and their associated loads. In some embodiments, it is preferable for the cells **40** to support loads in accordance with known AASHTO (American Association of State Highway and Transportation Officials) H-20 load requirements. In addition, the cells **40** are to be configured to be stackable side by side, as shown in FIG. 1, without interlocking or coupling together. That is, while adjacent cells **40** may contact each other and thus be subject to frictional forces therebetween, they are not otherwise coupled to each other. For example, adjacent cells **40** are not attached to each other, and cells **40** do not contain any features allowing them to be connected to any other cells **40**. In this manner, cells **40** can be relatively easily removed in the event that any features below or within them, such as

(32) utilities, service lines, or the like, must be accessed for maintenance or repair. Leaving cells **40** unconnected to each other also acts to localize the failures of any individual cells **40**, so that the failure of one cell **40** does not compromise any other cells **40**, and the failed cell **40** can be relatively easily replaced by simply removing it and adding another in its place. Finally, the various component parts of cells **40** may be designed to be injection-moldable, although embodiments of the invention contemplate any method of fabricating any part of the cells **40**.

(33) These constraints are satisfied by the structural cell design of FIG. 2A, which illustrates further details of the structural cells **40** of FIG. 1. In FIG. 2A, the structural cell **40** has a base or lower portion **100**, legs or vertical supports **110**, and a top, deck, or upper portion **120**. The base **100**, legs **110**, and deck **120** are fabricated separately and assembled to form the cell **40** shown.

(34) In the embodiment shown, the assembled cell **40** is generally rectangular, with three support members **110** along each of its longer sides for a total of six vertical supports **110**. More specifically, a vertical support **110** is located at each corner of the cell **40**, with two additional vertical supports **110** located inbetween. The lower frame **100** is also relatively thin and therefore pliable to a degree, so as to conform to small irregularities in the foundation **60**. It can be observed that the cell **40** leaves the volume within largely unobstructed, i.e., free of excessive numbers of support members, allowing large roots and other large-sized objects to be placed within. This yields significant advantages, as cells **40** not only contain relatively large amounts of open space, but this

open space is also easily accessible for penetration by roots (such as those of tree **10**) or other objects. Thus, not only is space available for roots and other objects, but they can also grow into, or be placed within, the cells in a relatively unimpeded fashion. Cells **40** can thus be used in connection with even very large trees with large root systems, as the cells **40** offer very little in the way of obstructions to impede the growth of even large roots therethrough. This creates cells **40** with relatively large open structures that can be easily accessed and thus, for example, are easily filled with soil, in contrast to cells with excessive numbers of support members that inhibit the placement of soil or other objects within.

(35) As can be seen in FIGS. 2B and 2C, each cell **40** has two rows of three legs **110** each, with three legs **110** positioned evenly along each longer side of cell **40**, and two legs **110** positioned at each end of each shorter side of cell **40**. As will be described further below, bottom ends of the legs **110** are inserted into recesses in the base **100**, and the deck **120** is applied to the top ends of the legs **110**.

(36) FIGS. 3A-3C illustrate further details of the base **100**. In this embodiment, the base **100** is a rectangular framelike structure with six recesses **200** interconnected by outer longitudinal supports **210** and inner longitudinal supports **230**, although any size and shape of base **100** is contemplated. The recesses **200** are each sized and shaped to accommodate a leg **110**, which can turn and lock into its recess **200**. Outer edges of the outer longitudinal supports **210** collectively form a perimeter around the base **100**, with inner longitudinal supports **230** also connecting adjacent recesses **200** while being located inside, and spaced apart from, outer longitudinal supports **210**. Outer edges **220** of recesses **200** protrude beyond the perimeter formed by the outer edges of the outer longitudinal supports **210**. That is, the recesses **200** are positioned to at least partially extend beyond the outer edges of outer longitudinal supports **210**.

(37) The base **100** also has crossbars **240** connecting diagonally-adjacent pairs of recesses **200**. That is, crossbars **240** connect recesses **200** at the corners of base **100** to recesses **200** located at the middle of the longer sides of base **100**. Pairs of crossbars **240** intersect, and footpads **250** are placed at their intersections. Footpads **250** are formed by placing curved supports **260** at the intersections of pairs of crossbars **240**, and can be used by workers when constructing the assemblies of cells **40** shown in FIG. 1. For example, footpads **250** allow workers safe places to step when the bases **100** are already placed below ground, so that the workers do not step on other more fragile parts of cells **40**. The footpads **250** are shown as circular and are sized to support a human foot in a work boot, although various embodiments contemplate any size and shape for the footpad **250**. Crossbar **250** is also located between the two recesses **200** placed at the middle of the longer sides of the base **100**, so as to add extra support in the vertical direction of FIG. 3A.

(38) It should be noted that the configuration of FIG. 3A yields significant savings in material over conventional structural cell bases. For example, as noted previously, the outer longitudinal supports **210** are set inward from the outer edges **220** of recesses **200**, resulting in less material used as compared to conventional structural cells whose outer supports extend parallel to their recesses (making them longer and thus using more material), while still maintaining structural integrity. Also, the present embodiment utilizes two supports **210**, **230** separated by a space inbetween, rather than a single wider, solid support. This also results in material savings. Likewise, the footpads **250** are configured as relatively thin outer curved supports **260** surrounding empty space, save for that portion of crossbars **240** extending within. The significant amount of empty space thus also provides a footpad made from a relatively small amount of material.

(39) The legs or vertical supports **110** are not limited to the length shown in FIG. 2A, but instead can be any length. In particular, one embodiment of the invention contemplates legs of two different lengths. In this manner, cells of at least three different heights can be made: a shorter cell with short legs, a medium-height cell with the longer legs, and a tall cell with each vertical support made of a longer leg connected to a shorter leg. This allows for the flexibility to make cells of different heights, so that the same component parts can be used to assemble cells suitable for many

different applications.

(40) FIGS. 4A-4C illustrate further details of a longer leg. Here, longer leg **300** includes ends with features **310** and **330** for connection to the base **100** and top/another leg respectively, as well as a central portion with longitudinal protrusions **320**. The features **310**, **330** are features allowing for turn and lock connection to the base **100** and top/other leg. For example, the features **310**, **330** can be slots and a corresponding pin or other extension sized to fit into the slot so that turning the leg secures the pin in the slot. The features **310**, **330** are configured to allow turn and lock connection to both the base **100** and another leg or the top **120**. For instance, the lower end of leg **300** has slots as features **310**, while the upper end of leg **300** has pins or extensions as features **330**, while the recesses **200** have pins or extensions, and the corresponding portions of the top **120** have snap fit joints. This allows the leg **300** to be turned and locked into place within a recess **200** of base **100**, whereupon either the lower end of another leg can be turned and locked onto the upper end of leg **300**, or the top **120** can be snapped into place on the upper end of leg **300**. If another leg is attached to the upper end of leg **300**, the top **120** is snapped onto the upper end of that added leg.

(41) The longitudinal protrusions **320** extend outward, adding to the radius of the leg **300** and thus improving the strength of the leg **300**. In particular, the protrusions **300** improve the bending stiffness and buckling strength of the leg **300**, as would be understood by one of ordinary skill in the art. The protrusions **320** can be oriented so that the center of each protrusion **320**, i.e. the point of maximum height or distance from the central axis of leg **300**, is oriented parallel to one of the sides of base **100**, i.e. in FIG. 3A, a line drawn through the centers of two opposing protrusions **320** of a leg **300** is oriented either horizontally or vertically, along either the longer side or the shorter side of the leg **300**. It has been found that this orientation provides desirable resistance to buckling. However, it should be noted that any number and orientation of protrusions **300** is contemplated. Also, the protrusions **320** can be of any cross-sectional shape. The shape of each protrusion **320** is not limited to the curved or domelike cross-sectional shape shown in FIG. 4C, but can be any cross-sectional shape that improves the strength of its leg **300**.

(42) FIGS. 5A-5C illustrate further details of a shorter leg. Similar to the longer leg **300**, shorter leg **400** has features **410**, **430** that are similar to the respective features **310**, **330** of longer leg **300**. That is, features **410**, **430** can be slots and pins/extensions respectively, which allow the shorter leg **400** to lock within recesses **200** of base **100**, and to be snapped onto the top **120**. Also, longitudinal protrusions **420** are shaped, and act, similar to protrusions **320** of leg **300**, increasing bending stiffness and buckling resistance.

(43) As noted above, cells **40** can utilize only the longer legs **300**, only the shorter legs **400**, a longer leg **300** combined with a shorter leg **400**, or a shorter leg **400** combined with another shorter leg **400**. This allows for cells **40** of at least four different heights.

(44) FIGS. 6A-6C illustrate the top or deck **120** in further detail. Here, deck **120** has recesses **500** corresponding in number and location to the recesses **200** of base **100**. That is, the deck **120** has six recesses **500** arranged in rectangular manner, with two rows of three recesses **500**. The deck **120** has a rectangular shape, with recesses **500** located at each corner and two recesses **500** located at the midpoints of each longer side, where the outer edges of recesses **500** each protrude beyond the sides **506** of deck **120**. However, the deck **120** is not limited to this configuration, and may have any shape, and any number or position of recesses **500**, so long as the recesses **500** are positioned to vertically align with the recesses **200** of base **100**. Indeed, the deck **120** need not be rectangular in shape, and need not be of the same general shape as that of the base **100**, even though the base **100** and deck **120** are shown as each having a rectangular shape in the present embodiment. The recesses **500** are holes that extend through the deck **120**, and contain features allowing for snap fit (or other suitable attachment) to legs **300**, **400**.

(45) The deck **120** also has stiffeners **502** arranged in diagonal manner between recesses **500**, as shown. The stiffeners **502** are bar- or beam-shaped structures placed within channels formed in the deck **120**, and connected to the deck **120** at their ends. The stiffeners **502** can be made of the same

material as the deck **120**, or may be made of a stronger material such as a steel. The stiffeners **502** can be integrally formed with the deck **120** or may be separate members attached to the deck **120**. Connection to the deck **120** can be made in any manner, such as by forming the stiffeners **502** integrally with the deck **120**, snap fitting the stiffeners **502** within appropriate features formed in the deck **120**, attaching the deck **120** and stiffeners **502** to pins that secure the deck **120** to the stiffeners **502**, or the like. Any connection method or apparatus is contemplated.

(46) The stiffeners **502** add to the strength and bending stiffness of the deck **120**. The body of the deck **120** is formed of a honeycomb structure **504** for sufficient strength while saving weight and material. It is contemplated that the cells **40** will often be filled with soil first, then the deck **120** attached. However, and in the alternative, hexagonal holes of the honeycomb structure **504** are sized to allow soil to relatively easily fall through the deck **120** and into the space within the cell **40**, allowing for the cells **40** to be filled with soil after the deck **120** is attached. For example, in one embodiment, the hexagonal holes measure 30 mm across the diagonal, and 27.3 mm across the flats. In this manner, the cell **40** can be easily filled with soil even after it is fully assembled, by simply pouring soil on top of the deck **120** and allowing it to fall through the hexagonal holes and into the cell **40** (or perhaps working the soil slightly to allow it to fall through the deck **120**). The hexagonal holes are not limited to this particular shape however, and embodiments of the invention contemplate any sizes and arrangements of holes that allow soil to readily fall therethrough.

(47) Embodiments of the deck **120** also have arched shapes, to better support loads. For example, the upper and lower surfaces of deck **120** are slightly curved or arched, as shown in FIGS. **6B** and **6C**. More specifically, the upper surface has an arch or curve that extends across the entire deck **120**, with lowest points at the two opposing shorter sides of the deck **120** and a highest point in the middle. Similarly, the lower surfaces are also curved or arched between adjacent recesses **500**, with each curve or arch having lowest points at or near two adjacent recesses **500**, and highest points inbetween. Other embodiments need not include these arches or curved surfaces, or may include only some of them. Likewise, the degree or amount by which each surface is arched may vary to any degree that allows the deck **120** to facilitate maintenance of root growth medium within cells **40** in a substantially uncompacted state.

(48) As described above, the cells **40** are modular and designed to be used with legs of multiple lengths, allowing for cells **40** of multiple heights for use in many different applications. FIGS. **2A-2C**, as already described, illustrate a cell **40** constructed using a single set of longer legs **300**. Meanwhile, FIGS. **7A-7C** illustrate a cell **40** constructed so that each vertical support is made up of a longer leg **300** and a shorter leg **400** connected to each other, and FIGS. **8A-8C** illustrate a cell **40** constructed using a single set of shorter legs **400**. The cells **40** of FIGS. **2A-2C** may be used in applications where an intermediate-sized cell **40** is desired, such as, for example, when an intermediate amount of uncompacted soil is desired, or when the particular site is sized so that it can only accommodate intermediate-sized cells **40**.

(49) In FIGS. **7A-7C**, each recess **200** has a longer leg **300** attached thereto, as above via a turn and lock mechanism or any other suitable mechanism. Attached to each longer leg **300** is a shorter leg **400**, again attached via a turn and lock mechanism or any other suitable mechanism. The shorter legs **400** are then attached to recesses **500** of the deck **120**, via snap fit or other appropriate mechanisms. The cells **40** of FIGS. **7A-7C** thus employ both a longer leg **300** and a shorter leg **400** for each vertical support, and are therefore taller than those of FIGS. **2A-2C** which employ only a single longer leg **300**. Accordingly, the cells **40** of FIGS. **7A-7C** may be used in connection with larger sites that can accommodate larger cells **40**, when a large amount of uncompacted soil is desired, more stormwater is desired to be retained, or the like.

(50) In FIGS. **8A-8C**, each recess **200** has only a shorter leg **400** attached thereto, again via a turn and lock mechanism or any other suitable mechanism. The shorter legs **400** are then attached to recesses **500** of the deck **120**, via snap fit or other appropriate mechanisms. The cells **40** of FIGS. **8A-8C** thus employ only a single shorter leg **400** for each vertical support, and are therefore shorter

than those of FIGS. 2A-2C. Accordingly, the cells **40** of FIGS. 8A-8C may be used in connection with smaller sites that can only accommodate these smaller cells **40**, when a relatively small amount of uncompacted soil is desired, less stormwater is desired to be retained, or the like.

(51) In operation, a foundation **60**, which is typically impermeable to tree roots, can be formed. Often, the foundation **60** is soil which has been compacted to at least 95% proctor to serve as an underlying foundation for construction, as is known. However, the foundation **60** can be any other layer that is highly resistant or effectively impermeable to tree roots, such as bedrock, concrete, aggregate, or the like. Then, a number of cells **40** are assembled and placed side by side around the rootball of a tree **10**, as shown in FIG. 1. The tree **10** may already be present, or the cells **40** may be arranged with a gap inbetween, and the rootball of a tree **10** may be inserted into the gap. The number of cells **40** used is preferably sufficient to provide enough volume of uncompacted soil to allow the roots of tree **10** to grow to their full natural capacity, although any number is contemplated. The cells **40** are not connected to each other. Rather, they are simply placed side by side, and not coupled to each other. Indeed, it can be seen that the individual cells **40** have no mechanism by which they can be attached or otherwise coupled to each other.

(52) The spacing between cells **40** may range. For example, depending on the application desired, adjacent cells **40** may be immediately adjacent to each other, and perhaps even contacting (touching but not otherwise coupled to) each other. At the other extreme, adjacent cells **40** may be placed several inches or more apart from each other. Any intermediate positions are also contemplated. As one nonlimiting example, adjacent cells **40** may be placed from 0 to 5 inches apart from each other. As above, this helps prevent the failure of one cell **40** from compromising its adjacent cells **40**, and also allows for easier repair of underlying utilities, service lines, or anything else placed within or under the cells **40**.

(53) Next, soil or any other tree root growth medium is poured into the cells **40** and evenly spread therethrough, after which the decks **120** are placed over the cells. In this manner, the cells **40** are filled with substantially uncompacted, or loosely compacted, soil. It may also be possible to attach the decks **120** first then pour soil/growth medium over the decks **120**, where it falls through (or can be readily manipulated so as to fall through) the holes in the decks **120** to fill the empty space within each cell **40**.

(54) Subsequently, a geotextile layer is typically placed upon the decks **120**, and the aggregate **30** and hardscape **20** are poured upon the geotextile layer. The weight of the aggregate **30** and hardscape **20** then acts to push the geotextile layer partially into the depressions **300**. This acts to secure the deck **120** and cells **40** against any lateral movement, adding to the structural stability of the cells **40**. Stability is further aided by the soil **50**, which also supports the cells **100** against any lateral movement.

(55) In the resulting configuration, one of ordinary skill in the art will observe that the cells **40** support the weight of their overlying aggregate **30**, hardscape **20**, and any traffic (foot or vehicle) that passes thereover. As the cells **40** support this weight, the tree growth medium is left substantially uncompacted, supporting only its own weight. Accordingly, the rootball of tree **10** is surrounded with a significant volume of substantially loosely compacted soil, which allows the roots of tree **10** to grow therethrough, providing space and a suitable medium for the roots to grow to their full natural size relatively unencumbered. This is in contrast to many conventional urban tree growth sites, which are surrounded by hardscape such as concrete roads and sidewalks that require their underlying earth to be packed so densely and solidly to support the hardscape, that tree roots cannot grow therethrough, thus stunting their growth as well as that of the tree.

(56) Thus, embodiments of the invention allow for growth of large trees in areas where they could not be grown before, in particular dense urban areas containing significant hardscape covering. This also allows for much more stormwater retention in these areas, which is a significant advantage over conventional dense urban areas, whose hardscape covering typically converts most stormwater to relatively useless runoff that also contributes to problems such as flooding and the

like. Instead of simply producing runoff, embodiments of the invention retain stormwater, using it to water trees and provide other benefits, while also reducing the problems commonly associated with runoff, such as flooding.

(57) Retention of stormwater also provides further benefits, such as pollutant removal and cleaning of runoff water. More specifically, both the loosely compacted soil and tree roots clean and/or filter water that they capture. Accordingly, instead of simply producing runoff that becomes contaminated as it picks up impurities and chemicals from the ground, embodiments of the invention retain and clean stormwater so that it can be used in tree or plant growth, etc.

(58) In terms of specific numbers, since the cells **40** support substantially the entire weight of overlying hardscape, aggregate, and commercial traffic, the soil or root growth medium does not experience any compressive force except that of its own weight. Accordingly, the cells **40** can maintain the soil or root growth medium in an uncompacted state, e.g., at approximately 80% proctor or less (roughly corresponding to compaction by human foot, or less). In contrast, the hard-packed soil created by general construction is typically 90-95% proctor or so, which is known to be effectively impenetrable to tree roots. Trees planted in typical construction sites, such as under pavement supported by hard-packed soil, commonly only have a small area in which to grow roots (usually a small volume dug out from the hard-packed soil of a construction area), resulting in a stunted root system and a smaller, less healthy tree. In contrast, the cells **40** of embodiments of the invention allow for large areas of substantially uncompacted, or loosely compacted, soil to exist even under or next to construction sites and other areas of hard-packed soil, resulting in growth of large, healthy trees even in areas that could not conventionally support such trees, like dense urban areas covered by concrete or asphalt.

(59) As above, such cells **40** are largely open, indeed having up to 90% or more of their volume being free volume available to be taken up by root growth. The significant amount of open area within cells **40** also allows for other structures to extend therethrough. For example, water pipes, electrical or other lines may be run into or through the cells **40**, so that utilities need not be routed around any system of cells **40**.

(60) Attention now turns to further features of the cells **40** of embodiments of the invention. In some though not all applications, it is desirable for the bases **100** and decks **120** to be stackable upon each other, so that the bases **100** and decks **120** can be more readily and more efficiently transported. FIGS. **9A-9C** illustrate stacked bases **100** configured according to an embodiment of the present invention. Here, the recesses **200** of each base **100** have rims **600** extending upward, out of the main body of their base **100**. The rims **600** can extend upward to any distance (i.e. each can have any height), but in one embodiment can be sized to provide protection to the joints between their recesses **200** and legs **300/400**, as well as provide sufficient clearance between stacked bases **100**, without providing excessively large clearance between stacked bases **100** and thus reducing the efficiency and compactness with which they can be stacked.

(61) Each recess **200** can also have a corresponding circular depression or opening **610** formed on the other side of the base **100** as the rim **600**, and sized to hold the rim **600** of another base **100**. In this manner, the rims **600** of one base **100** may engage with the openings **610** of another base **100** that is stacked on top, holding it securely in place. Thus, stacked bases **100** can be coupled together in lateral directions (i.e. kept from sliding laterally off of each other, and thus remaining stacked), but substantially uncoupled in the vertical direction. That is, the base **100** at the top of the stack is kept from sliding laterally off of the stack, but is free to be removed from the stack simply by lifting it off. The recesses **200** may or may not be present, and the invention includes embodiments in which they are, as well as embodiments in which they are not. In high loading situations where recesses **200** are desired, the recesses **200** and surrounding material of the base **100** may be reinforced, such as by thickening the walls of these structures, adding reinforcement, or the like.

(62) Embodiments of the present invention may also include stackable decks **120**. FIGS. **10A-10C** illustrate stacked decks **120** configured according to an embodiment of the present invention. Here,

the recesses **500** have rims **700** extending downward, away from the main body of their deck **120**. The rims **700** can extend downward any distance, i.e. each can have any length, but in one embodiment can be sized to provide protection for the joints between recesses **500** and the tops of legs **300/400**, as well as provide sufficient clearance between stacked decks **120**. Conversely, the rims **700** are made short enough to avoid excessively large clearance between stacked decks **120**, which would reduce the efficiency and compactness with which they can be stacked.

(63) Each recess **500** can also have a corresponding circular depression or opening **710** formed on the upper surface of the deck **120** (i.e. on the other side of the deck **120** as the rim **700**). The rims **700** of one deck **120** may thus engage with the openings **710** of another deck **120** that is stacked below, thus holding it securely in place. Like the bases **100**, the decks **120** when stacked can thus be coupled together in lateral directions (i.e. kept from sliding laterally off of each other, and thus remaining stacked), but substantially uncoupled in the vertical direction. That is, the deck **120** at the top of the stack is kept from sliding laterally off of the stack, but is free to be removed from the stack simply by lifting it off.

(64) It should be noted that the above described cells **40** and their various components may be made of any materials, and may be of any specific dimensions, that allow the cells **40** to support overlying hardscape and its commercial vehicle traffic while maintaining the soil within the cells **40** in a substantially uncompacted state and allowing for relatively unimpeded tree root growth therethrough. That is, the cells **40** must be sufficient to support the weight of the overlying hardscape and commercial vehicle traffic by themselves, while transferring little to none of this weight to the soil within, while also being large and open enough to allow for relatively unimpeded growth of tree roots within. Any such materials and dimensions that allow for this to happen are contemplated.

(65) As one nonlimiting example, the various components of cells **40** may have certain specific dimensions as follows. The base **100** may measure 600×1200 mm, and the height as measured from the bottom of base **100** to the top of a rim **600** may be 110 mm. The deck **120** may have similar measurements (in one case, 592×1193×107 mm). Longer legs **300** may be 725 mm in length, 160 mm in diameter at their lower end, 158 mm in diameter at their upper end, and 153.7 mm in diameter at their central portion and as measured from the center of one longitudinal protrusion **320** to its opposite protrusion **320**. The nominal wall thickness of longer legs **300** may be 3.175 mm, and the upper and lower portions may each be approximately 100 mm in length. Similarly, shorter legs **400** may be 365 mm in length, with the remaining dimensions being the same as those of longer leg **300**.

(66) One of ordinary skill in the art will realize that, for these and other dimensions, the base **100**, longer legs **300**, and shorter legs **400** may be made of injection molded high density polyethylene (HDPE), with the deck **120** made of a stronger material like a glass-filled resin. This glass-filled resin may be a 30% glass-filled polypropylene, or any other material of similar or greater strength. This is in contrast to prior designs such as those of U.S. Pat. Nos. 7,080,480 and 8,065,831, which both utilized 30% glass-filled polypropylene or stronger material for their entire cells, rather than the cheaper HDPE. Accordingly, embodiments of the present invention allow for significant cost savings in materials, as cheaper HDPE is used for a large portion of the cells **40**. One of ordinary skill in the art will also realize that the invention is not limited to these dimensions or these materials, and instead contemplates any dimensions, and any corresponding materials, that lead to a cell that is capable of supporting overlying hardscape and its commercial vehicle traffic while maintaining the soil within in a substantially uncompacted state and allowing for relatively unimpeded tree root growth therethrough.

(67) The above and other embodiments are configured to support the weight of overlying hardscape (e.g. a standard concrete or asphalt roadway, pavers, etc.) and aggregate that cover the entire upper surface of the cell, as well as commercial vehicle traffic (e.g., trucks, cars, emergency vehicles, and any other vehicles commonly found on roadways) thereover, to a degree at least equivalent to an H-

20 loading condition. For example, the above and other embodiments can support a load of at least 21.0 psi when the hardscape **20** is pavers with concrete, and at least 15 psi when the hardscape **20** is just pavers.

(68) As noted above, one notable advantageous application of cells **40** is stormwater management. As one example, cells **40** can be used to create a ponding area for storing rainwater runoff. In such an application, a number of cells **40** may be placed (not necessarily side by side) under hardscape, and may or may not be filled with soil. Trees **10** also may or may not be present. Ingress and egress for water are provided, such as a runoff drain or simply an open area in the overlying hardscape, as well as an underdrain that may extend into the volume outlined by cells **40** to convey excess water somewhere else, such as to a primary stormwater system.

(69) Another application of cells **40** is that of underground bioretention systems. A bioretention system, sometimes referred to as a rain garden, is typically a volume of soil and/or other materials used to collect and hold runoff water, and remove pollutants therefrom. Stormwater is typically collected into a treatment area, which is an area of ground containing different layers such as a grass strip, sand bed, one or more organic layers, soil, and plants.

(70) These layers collect the runoff water, filter and aerate it, support the growth of microorganisms which break down certain undesired compounds in the runoff water such as hydrocarbons, and absorb other undesired compounds. The “cleaned” runoff water then gradually propagates into the soil surrounding the bioretention system, evaporates away, or can be removed for use elsewhere. The cells **40** can create such a bioretention system underground. For example, the configuration of FIG. **1** can be filled not with solely conventional soil as in some above-described applications, but with layers that constitute a bioretention system. As above, such layers can include a sand bed, one or more organic layers, soil of any type, or the like. Embodiments of the invention encompass any material layers suitable for use in a bioretention system. Ingress and egress for runoff water may also be added if desired, such as drains or pipes that convey water into and out of the cells **40**.

(71) Runoff water is thus channeled into the cells **40**, where the various layers as well as the roots of tree **10** filter, aerate, and break down pollutants in the water. The water is then taken up by the tree **10**, evaporates, enters the surrounding soil, or is removed by the water egress. In this manner, cells **40** of embodiments of the invention can be deployed to create an underground bioretention system that serves the dual purpose of supporting tree growth and has the added advantage of utilizing the tree and its root system to improve the function of the system.

(72) It should be noted that bioretention systems formed utilizing embodiments of the present invention offer significant advantages over conventional bioretention systems. For example, conventional bioretention systems are known to degrade over time. In contrast, it has been found that bioretention systems constructed according to embodiments of the present invention actually improve over time. In particular, as the trees within it grow larger and more mature root systems, the larger root system absorbs more water and removes more pollutants. It has also unexpectedly been found that the benefit imparted to bioretention systems by incorporating trees within is a greater than linear benefit. That is, such systems improve in greater than linear manner as a function of tree growth, with larger trees offering much more benefit (water absorption and pollutant absorption/breakdown) than smaller ones.

(73) It should further be noted that these above mentioned advantageous embodiments all come with the further benefit of more efficient land use. The structural cells **40** are all sufficient to support conventional hardscape above. Thus, all of the above advantageous embodiments can be implemented belowground without compromising any of the conventional structures or systems aboveground. For example, bioretention systems, stormwater retention systems, and the like can all be implemented underneath sidewalks and streets, so that they do not reduce the amount of parking spaces available. Larger trees can be grown without reducing the amount of sidewalk available for walking. In this manner, the above described advantages can be accomplished without reducing the aboveground real estate available for conventional uses.

(74) The foregoing description, for purposes of explanation, used specific nomenclature to provide a thorough understanding of the invention. However, it will be apparent to one skilled in the art that the specific details are not required in order to practice the invention. Thus, the foregoing descriptions of specific embodiments of the present invention are presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. For example, the cells **40** can take on any shape besides rectangular, and can have any number of vertical supports, each of which can contain any number and combination of legs. The cells **40** can also be made of any material or materials sufficient to allow cells **40** to support overlying aggregate and hardscape as well as traffic loading. The embodiments were chosen and described in order to best explain the principles of the invention and its practical applications, to thereby enable others skilled in the art to best utilize the invention and various embodiments with various modifications as are suited to the particular use contemplated. Additionally, different features of the various embodiments of the present invention, disclosed or otherwise, can be mixed and matched or otherwise combined so as to create further embodiments contemplated by the invention.

Claims

1. A structural cell system for supporting an underground bioretention system to collect and remove pollutants from runoff storm water, the system comprising: a base comprising a plurality of receptacles and a plurality of support members interconnecting the receptacles; a plurality of legs each sized and shaped to be attachable to the base within one of the receptacles so as to extend from the base, and to be attachable to another of the legs so that pairs of legs attached to each other collectively extend from the base; a top attachable to the legs via a plurality of recesses, wherein the top comprises a honeycomb structure and comprises a plurality of channels arranged in a diagonal intersecting manner between the plurality of recesses, each channel comprising a bar- or beam-shaped stiffener formed therein, the stiffeners intersecting each other in a diagonal manner; wherein the base, the top, and the legs attached thereto define a volume comprising a treatment area comprising a plurality of layers of different material to collect, filter and aerate runoff water, the plurality of layers comprising: a first layer comprising soil; a second layer comprising at least one of a grass strip, plant material, or an inorganic material, wherein the plurality of layers support the growth of microorganisms which break down and thereby remove pollutants from the runoff water.
2. The structural cell system of claim 1, further comprising one or more ingress or egress to convey runoff storm water into and out of the cells.
3. The structural cell system of claim 2, wherein the one or more ingress or egress comprise drains or pipes that convey runoff storm water into and out of the cells.
4. The structural cell system of claim 2, wherein the one or more ingress or egress is configured to enable the cleaned runoff water to propagate into the soil surrounding the bioretention system, evaporate away, or be removed for use elsewhere.
5. The structural cell system of claim 4, wherein the soil surrounding the bioretention system further comprises tree roots therein and enables water to be taken up by the tree roots.
6. The structural cell system of claim 5, wherein the cell is capable of supporting overlying hardscape and commercial vehicle traffic while maintaining the soil within in a substantially uncompacted state and allowing for relatively unimpeded tree root growth therethrough.
7. A structural cell system for supporting an underground bioretention system to collect and remove pollutants from runoff storm water, the system comprising: a base comprising a plurality of receptacles and a plurality of support members interconnecting the receptacles, wherein each of the plurality of receptacles extends at least partially beyond outermost edges of the plurality of support members; a plurality of legs; a top attachable to the legs; wherein the base, the top, and the legs

attached thereto define a volume comprising a treatment area comprising a plurality of layers of different material to collect, filter and aerate runoff water, the plurality of layers comprising: a first layer comprising soil; a second layer comprising at least one of a grass strip, plant material, or an inorganic material, wherein the plurality of layers support the growth of microorganisms which break down and thereby remove pollutants from the runoff water.

8. A structural cell system for supporting an underground bioretention system to collect and remove pollutants from runoff storm water, the system comprising: a base comprising a plurality of receptacles and a plurality of support members interconnecting the receptacles; a plurality of legs each sized and shaped to be attachable to the base within one of the receptacles so as to extend from the base, and to be attachable to another of the legs so that pairs of legs attached to each other collectively extend from the base; a top attachable to the legs via a plurality of recesses, wherein the top comprises a honeycomb structure and comprises a plurality of stiffeners arranged in a diagonal intersecting manner between the plurality of recesses, wherein each of the stiffeners comprises a material different than a material of the honeycomb structure; wherein the base, the top, and the legs attached thereto define a volume comprising a treatment area comprising a plurality of layers of different material to collect, filter and aerate runoff water, the plurality of layers comprising: a first layer comprising soil; a second layer comprising at least one of a grass strip, plant material, or an inorganic material, wherein the plurality of layers support the growth of microorganisms which break down and thereby remove pollutants from the runoff water.
